Stellar Temperature Inference
Project Guide for 5B

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Mathematical Methods for Physicists II

Python Notebook Guide

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This guide follows the simulated blackbody fit in 5B. Read the Bayesian foundations guide for the shared statistics; the emphasis here is on spectral shape and the temperature–brightness trade-off.

1 What stellar temperature means

A star has no single physical temperature throughout. Its effective temperature is defined by the total energy radiated per unit surface area:

$$F_{\text{surface}} = \frac{L}{4\pi R^2} = \sigma_{\text{SB}} T_{\text{eff}}^4.$$

Here L is luminosity over all wavelengths and R is the stellar radius. For the simulated blackbody, the fitted temperature is the effective temperature by construction. For a real star, a temperature fitted from continuum colour alone need not equal the effective temperature: absorption, dust, or instrumental response can change the spectral shape. The free brightness scale also means the notebook does not infer radius or distance separately.

2 Why temperature changes the shape

The model uses Planck's radiance per unit wavelength,

$$B_\lambda(T) = \frac{2hc^2}{\lambda^5} \left[\exp\left(\frac{hc}{\lambda k_{\text{B}} T} \right) - 1 \right]^{-1}.$$

The code converts nanometres to metres. The exponent is dimensionless, and `expm1` evaluates the denominator accurately for small exponents. Dividing by a fixed reference radiance at 550 nm and 5800 K sets the units and does not constrain the fit. A free multiplier A accounts for overall brightness, including geometric scaling.

Wien's law gives $\lambda_{\text{max}} T \simeq 2.898 \times 10^{-3}$ m K, so the wavelength peak is near 500 nm at 5800 K. This is the peak in wavelength radiance, not the wavelength corresponding to the peak in frequency radiance. Temperature changes the peak and shape; A scales the curve.

At sufficiently long wavelengths, the Rayleigh–Jeans approximation is $B_\lambda \simeq 2ck_{\text{B}}T/\lambda^4$. The normalization then mainly constrains AT, so a higher temperature can be offset by lower brightness. The chosen infrared band is not fully in this limit. The fit uses full Planck radiance; the approximation explains the direction of the trade-off.

3 Walk through the computation

Simulate a 5800 K blackbody with $A = 1$ and independent Gaussian noise of the same absolute scale at each wavelength. The first plot shows the measurements and their generating curve. Each fit uses a Boolean mask to select data while retaining the same priors and noise scale.

For each temperature, evaluate spectral values $s_i(T)$. Expand the squared residuals to avoid constructing a three-dimensional array:

$$\sigma^2 \chi^2(T, A) = \sum_i y_i^2 - 2A \sum_i s_i(T) y_i + A^2 \sum_i s_i(T)^2.$$

Compute the three sums before varying A, then use broadcasting to evaluate the quadratic. Uniform priors span 3500–8500 K and brightness 0.5–1.5. After normalization, sum over brightness to obtain temperature mass and divide by cell width to plot its density.

4 What changes when wavelengths are removed?

Selection	Points	Mean temperature (K)	Central 95% interval (K)
Full (250–3000 nm)	100	5805	[5736, 5875]
Visible (400–700 nm)	11	5724	[5546, 5909]
Infrared (1300–2800 nm)	54	6272	[4875, 8249]

The infrared result is much broader in this run despite using more points than the visible fit. Tail measurements mostly constrain a combination of parameters. The selections also differ in sample count and signal relative to noise, so this is an illustration rather than a controlled comparison of wavelength alone.

The infrared joint plot shows a ridge from cooler and brighter to hotter and dimmer. The full-range plot has a compact region. Compare spread and shape because the panels use separate colour scales. The red cross marks the simulated values. The infrared result extends toward the prior limits, making a wider-prior check useful.

5 Read the residuals

The final overlay uses the posterior mean temperature and brightness. Residuals $(y_i - m_i)/\sigma$ have RMS about 1.02. Scatter around zero without a clear trend is consistent with the assumed noise for this run. The error bars describe measurement scatter. A band for uncertainty in the fitted spectrum would propagate the joint posterior through the prediction.

6 Try it yourself

1. Change the generating brightness and check that both the generating curve and measurements use it.
2. Compare equally sized selections and repeat across noise draws to separate wavelength effects from sample size and noise realization.
3. Widen one prior bound at a time and inspect the infrared ridge and interval.

Real spectra can include absorption lines, extinction, calibration effects, and wavelength-dependent or correlated noise. These intervals are conditional on the simplified continuum and noise model.

7 Further reading

OpenStax, *University Physics, Volume 3*, section 6.1, Blackbody Radiation. For the stellar definition, see Swinburne Astronomy Online, COSMOS, Effective Temperature.